batteries every year, which contain hazardous chemicals. These batteries are not properly disposed of, so they pose big threat to the environment and health.

The author's prototype of a clock power supply using solar PV and a supercapacitor is shown in Fig. 1. The aim is to run the clock non-stop for at least 15 years under all weather conditions. This will eliminate the use of about 15 AA cells for each clock and reduce hazardous chemical waste.

Fig. 2 shows circuit diagram of the proposed power supply. The circuit uses two small 6V/100mA (PV1 and PV2) solar panels, two 1N5819 Schottky diodes (D1 and D2), three 1N4007 rectifier diodes (D3, D4, and D5), shunt regulator ICTL431 (IC1), 100F/2.7V supercapacitor (C1), 4700µF/10V electrolytic capacitor (C2), and a few other components.

Following are the specifications of the solar panels:

Voltage at maximum power = Vmp = 5V (depends on sunlight intensity)

Current at maximum power = Imp = 100mA

Open circuit voltage = Voc = 6V

Size = 70mm × 70mm

These panels are connected in parallel using Schottky diodes D1 and D2. The diodes block current from one panel entering the other panel.

The panel output voltage Vpv is fed to supercapacitor C1 through current-limiting resistor R1. Voltage regulator IC1 is connected across C1 to regulate the capacitor's voltage to 2.5V.

IC1 ensures that the capacitor voltage never exceeds the rated voltage of 2.7V. Note that, diodes D1 and D2 also stop C1 from discharging into the panels when there is no sunlight.

The output of capacitor C1 is connected to the clock terminals through current-limiting resistor R2. Across these terminals diodes D2, D3, and D4 are connected in series.

Total forward voltage of D2, D3, D4 = 0.65V × 3 = 1.95V

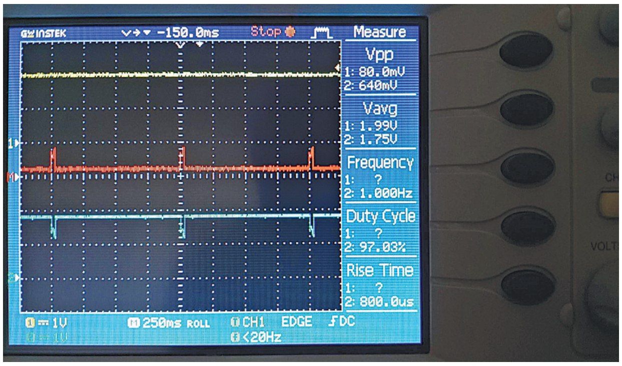

Fig. 3: Voltage waveforms (Test conditions: R2 = 100Ω; C2 not mounted) CH1 (Yellow Trace)→Capacitor voltage (C1); CH2 (Blue Trace)→Voltage +Vclock:M (Red Trace)→Ch1–Ch2

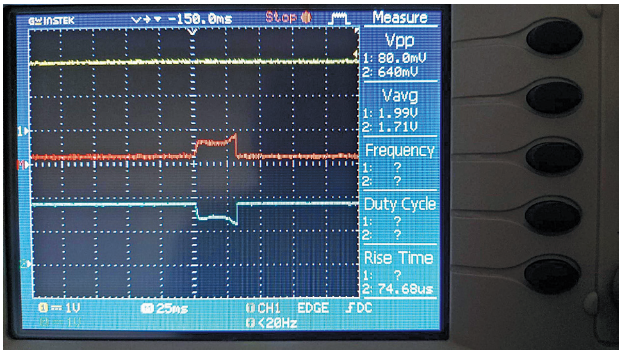

Fig. 4: Voltage waveforms expanded with test conditions: R2=100Ω; C2 not mounted; CH1 (Yellow Trace)→Capacitor voltage (C1); CH2 (Blue Trace)→Voltage +Vclock:M (Red Trace)→Ch1–Ch2

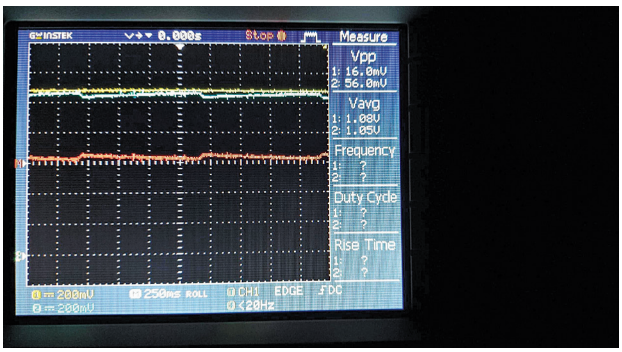

Fig. 5: Filtered voltage waveforms with test conditions: R2=220Ω; C2 mounted; CH1 (Yellow Trace)→Capacitor voltage (C1); CH2 (Blue Trace) →Voltage+Vclock:M (Red Trace)→Ch1–Ch2

The clock movement operates in the range of 2V to 1V. Thus, the three diodes limit the voltage Vclock to less than 2V. Capacitor C2 is used to reduce voltage dip whenever the movement draws pulse current.

The working of the project will be clearer from various waveforms shown in Fig. 3 through Fig. 5.

Fig. 3 shows C1 capacitor voltage and +Vclock voltage. These waveforms were captured without mounting the filter capacitor C2. Also, value of resistor R2 was changed to 100Ω.

As seen in Fig. 3, the capacitor voltage is stable at 1.99V. We see dip in +Vclock voltage every second when the clock movement draws pulse current.

Using math function of the oscilloscope, trace M (red) has been plotted, which gives difference between Ch1 and Ch2. This waveform is the voltage across resistor R2. Fig. 4 shows the same waveform expanded.

From the red trace of Fig. 4, we can calculate current through R2.

Steady current through R2 and D3, D4, D5 = 0.2V/100Ω = 2mA

Average pulse current = 0.6V/ 100Ω = 6mA

Actual pulse current drawn by the clock movement = 6mA–2mA = 4mA

Pulse duration = 30 milliseconds (ms)

Average current drawn by clock movement = 4mA × 30ms/1000 ms = 0.12mA

Average power drawn by the